

TASK 15 – PANDAS DATAFRAME BASICS

Veda Technology – Data Science Internship

1. Introduction

Pandas is one of the most widely used Python libraries for data analysis. Its primary data structure, the DataFrame, allows data to be stored and analyzed in a tabular format consisting of rows and columns.

This task focuses on understanding the basic operations performed on Pandas DataFrames.

2. Objective

The objectives of this task were to:

- Create a Pandas DataFrame manually.
 - Load a real dataset into a DataFrame.
 - Explore rows and columns.
 - Understand the shape and size of a DataFrame.
 - Check column names and data types.
 - Inspect DataFrame information.
 - Generate summary statistics.
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3. Tools and Technologies

- Python
 - Pandas
 - Scikit-learn
 - Jupyter Notebook / Google Colab
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4. Manual DataFrame Creation

A DataFrame was first created manually using a Python dictionary.

The student dataset contained the following columns:

- Name
- Age
- Marks
- Department

This demonstrated how structured data can be converted into a Pandas DataFrame using `pd.DataFrame()`.

5. Real Dataset

The Iris dataset was used as the real dataset for this task.

It was loaded using Scikit-learn and converted into a Pandas DataFrame.

The resulting dataset contains:

- 150 rows
- 5 columns

The columns include four numerical features and one categorical species column.

6. Data Exploration

Several Pandas functions and properties were used to explore the dataset.

`head()`

The `head()` function was used to display the first five rows of the DataFrame.

`shape`

The `shape` property was used to identify the number of rows and columns.

The Iris dataset has a shape of:

(150, 5)

This means there are 150 rows and 5 columns.

`columns`

The `columns` property was used to display the names of all columns.

`dtypes`

The `dtypes` property was used to identify the data type of each column.

`info()`

The `info()` function was used to inspect the overall structure of the DataFrame, including the number of entries, non-null values, data types, and memory usage.

`describe()`

The `describe()` function was used to generate summary statistics for numerical columns, including count, mean, standard deviation, minimum, maximum, and percentile values.

7. Shape and Size

The difference between `shape` and `size` was also explored.

For the Iris DataFrame:

Shape = (150, 5)

Therefore:

Size = 150 × 5 = 750 elements

`shape` describes the dimensions of the DataFrame, whereas `size` represents the total number of elements.

8. Interview Questions

What is a Pandas DataFrame?

A Pandas DataFrame is a two-dimensional labeled data structure used to store and analyze tabular data in Python.

What information does `info()` provide?

The `info()` function provides information about the DataFrame structure, including row count, column names, non-null values, data types, and memory usage.

What is the difference between `shape` and `size`?

`shape` returns the number of rows and columns, while `size` returns the total number of elements in the DataFrame.

For example, a DataFrame with 150 rows and 5 columns has a shape of `(150, 5)` and a size of 750.

9. Conclusion

The task successfully demonstrated the basic use of Pandas DataFrames for tabular data analysis.

The practical work included manual DataFrame creation, loading a real dataset, inspecting rows and columns, checking data types and structure, and generating summary statistics.

This task helped strengthen the foundation required for practical data analysis using Python and Pandas.